

Accelerate Boltzmann Sampling of Multimodal Distributions by Training with Log-Likelihood Ratio Discrepancy

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May 12, 2026

1 Introduction

The Boltzmann generator based on normalizing flows (Noé et al., 2019; Wu et al., 2020) has been a crucial approach to sampling complicated distributions, but on multimodal targets normalizing flows trained with the reverse KL are prone to mode collapse, where the modes lost during training corrupt the sample quality. To address this issue, several approaches have been proposed to mitigate mode collapse during training.

Suppose μ is the target distribution on $\mathbb{R}^d$ and ν is the pushforward distribution of the trained flow. The first approach is to replace the mode-seeking reverse KL by the mass-covering forward KL

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (1)$$

which is more stable on multimodal targets. Although the forward KL inherently requires samples from the target distribution, approximate target samples can be efficiently generated by Annealed Importance Sampling (AIS), which uses a sequence of reweighting and resampling steps to transport the pushforward distribution $\nu(x)$ to the target distribution $\mu(x)$.

Another representative method is the Flow AIS Bootstrap (FAB) (Midgley et al., 2022), which uses the α -divergence with $\alpha = 2$,

$$D_2(\mu\|\nu) = \int_{\mathbb{R}^d} \frac{\mu^2(x)}{\nu(x)} dx, \quad (2)$$

which remains purely energy-based and is capable of discovering new modes. Since the forward KL can be viewed as the limit of the α -divergence as $\alpha \rightarrow 1$, FAB can be seen as a generalization of the forward KL to a different choice of α . Compared to the forward KL, FAB requires importance sampling against the more delicate distribution μ^2/ν , which may cause training stability issues.

Although these approaches are theoretically capable of mitigating mode collapse, their training speed and stability are still limited by the complexity of the potential landscape. In many cases we already know the secondary modes of the target distribution μ , for example the multiple configurations of a molecule; can we integrate these known modes into the flow training?

In this paper we propose a new functional, the Log-Likelihood Ratio Discrepancy, defined by

$$\text{XX}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (3)$$

The shorthand XX denotes this nonnegative functional and is used purely for notational convenience.

References

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